ENGINEERING PORTFOLIO · 2026

ASHER STRAUS

Manufacturing & Design Engineer

I design parts and the processes that make them — moving between CAD, simulation and the shop floor, and testing until the model and the real thing agree.

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Manufacturing & Design Engineer

I am a Manufacturing and Design Engineering student at Northwestern, with a minor in Art and certificates in Segal Design and Robotics. I started wrenching on kart engines at 13, and that instinct still shapes how I work: model it, build it, find where the two disagree, and fix the real one. The projects in this book span fixture and production design, vehicle systems, simulation-driven structures, medical-equipment reliability, and manufacturing research.

EDUCATION

Northwestern University — Sept 2023 – Mar 2028 (expected)
McCormick School of Engineering · Evanston, IL

B.S. Manufacturing & Design Engineering
Minor in Art · Segal Design Certificate · Robotics Certificate
GPA 3.6 / 4.0 cumulative · 3.9 / 4.0 since sophomore year

EXPERIENCE

Siemens Healthineers — Jun 2026 – Present
Mechanical Engineering Co-op, Molecular Imaging

CAD models and drawings for SPECT and PET scanner components — resolving interferences, applying GD&T, and designing fixtures and brackets in Siemens NX. Contributing to a multi-axis stage for the SPECT research team.

Advanced Intelligent Manufacturing Lab — Nov 2025 – Apr 2026
Researcher · NSF ERC-HAMMER · Northwestern

Co-author on the MSEC 2026 / NAMRC54 poster for Smart Sheet Smith. Built the evaluation corpus and an automated drawing-generation pipeline; fabricated flexible tactile sensors.

P1 Motor Club — Apr 2023 – Present
Virtual Test Driver, Research & Promotions Intern

Test new track layouts in simulation and feed performance findings back to the engineers shaping the design.

MIT Beaverworks Summer Institute — Jun – Aug 2024
Teaching Assistant, "Basics of ASICs"

RELEVANT COURSEWORK

Manufacturing Engineering Design (DFM) · Computer Integrated Manufacturing · Geometry & Manufacturing · Mechanical Design & Manufacturing · Materials Selection · Mechatronics · Mechanics of Materials · Electronics Design · Human-Centered Product Design · Optimization

CLUBS

Northwestern Formula Racing (FSAE) · Baja SAE · Combat Robotics · SME Student Chapter

CAPABILITIES

Design & CAD
Siemens NX · SolidWorks · Onshape · AutoCAD / Civil 3D · GD&T · engineering drawings · fixture & jig design · DFM / DFR · snap-fit and compliant design

Analysis
ANSYS FEA · topology optimization · thermal simulation · MATLAB · Python · FMEA · cost modeling · lean / value stream mapping · line balancing

Fabrication
CNC, mill & lathe · waterjet · laser cutting · 3D printing (FDM, SLS, PETG) · welding · brazing · sand & iron casting · sheet metal · metrology

IN THIS PORTFOLIO

Bionic Wrench Manufacturing — Fixture design & production planning — 03

Formula SAE Brake System — Vehicle systems, simulation & machining — 04

Water Guard — Medical equipment reliability — 05

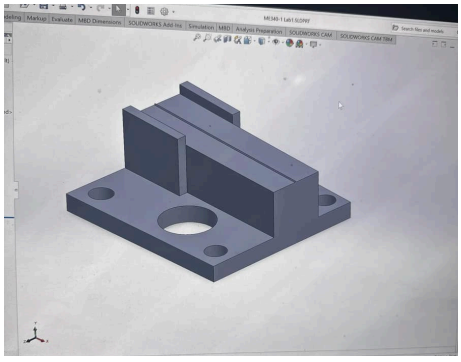
Smart Sheet Smith — Manufacturing AI research — 06

Bicycle Brake Caliper — Simulation-driven structural design — 07

ExtendIt — Human-centered product development — 08

Hands-On Capability — Fabrication & lab work — 09

Selected Additional Work — Research, vehicles & shop projects — 10



Part modelling and drawings in CAD



Brazing a copper assembly in the shop



Northwestern Baja SAE off-road racer

Bionic Wrench Manufacturing

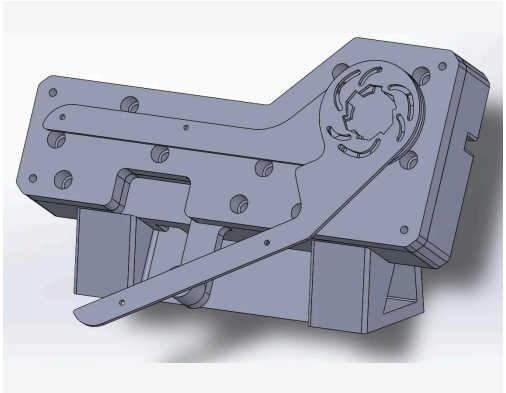
DSGN 386 — Manufacturing Engineering & Design · Northwestern · with Peter Wang

Reverse-engineered a commercial Bionic Wrench, then built everything needed to manufacture it: a seven-iteration assembly fixture, a documented single-operator line, and a production plan scaled to one million units a year — proven in a live 30-minute timed trial.

7	3:58	1M	$10.22
FIXTURE ITERATIONS	CYCLE TIME PER UNIT	UNITS / YEAR PLAN	COST PER UNIT

WHAT I DID

- Led the track geometry across all seven fixture versions and drove the metrology and dimensioned CAD drawing set.
- Dissected and dimensioned every component with calipers, micrometers and an optical comparator to infer fits, assembly order and the likely process for each part.
- Mapped the manufacturing and assembly flows. The maps exposed jaw riveting as the bottleneck and showed a two-worker layout left the second operator idle — so we consolidated to one.
- Wrote the SOP, FMEA and value stream map, and balanced a three-station single-operator line to 246 s per wrench.
- Sized the high-volume plan: 6 s takt at peak demand, 25 parallel U-shaped cells, ~$1.2M line investment, and a sub-five-minute pin-plate swap between jaw variants.



V7 — the final fixture, with the support that braces the inner plates when the wrench is fully open.

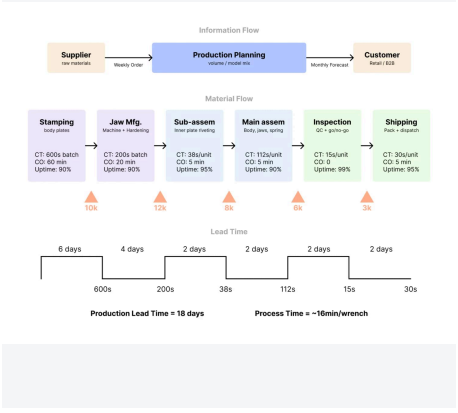
RESULT The timed trial produced four good wrenches at a 3:58 average cycle time, matching the planned rate. Both rejects traced to out-of-spec incoming jaws rather than to the process.



The commercial wrench we reverse-engineered.



The single-operator line: small riveter, assembly bench, large riveter.



Value stream map — 18-day lead time against ~16 min of process time.

TOOLS SolidWorks · Metrology · Waterjet · 3D printing · Lean / VSM · FMEA · Line balancing · Cost modeling

Formula SAE Brake System

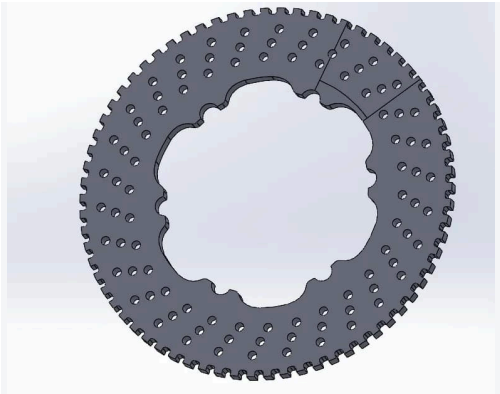
Northwestern Formula Racing — Suspension Member

Worked across the braking system, from a hand calculation of the pedal force needed for a 1 G stop through ANSYS structural and thermal verification — then machined the rotors, routed the lines and bled the system on the car.

1 G	690 lbf	42 lbf	~16×
HAND-CHECK STOP	TIRE BRAKING FORCE	DRIVER PEDAL FORCE	SYSTEM GAIN

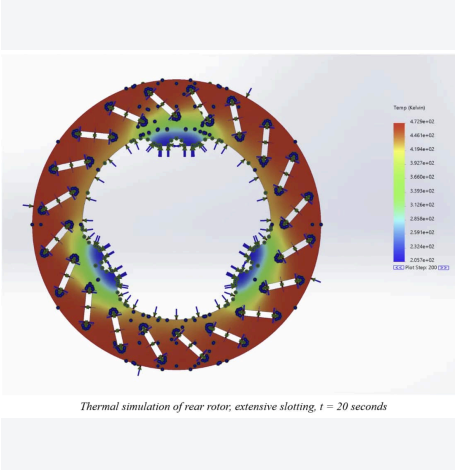
WHAT I DID

- Built the analytical model of the full force chain — pedal ratio, master cylinder pressure, caliper clamp force, pad friction, rotor torque, tire force, vehicle deceleration.
- Improved the team's MATLAB brake simulation so rotor diameter, master cylinder bore and pedal ratio could be swept quickly, and tuned the front/rear bias to cross over without rear lockup.
- Designed the rotor in CAD, balancing effective braking radius and thermal mass against rotational inertia and packaging inside the wheel.
- Verified in ANSYS: stress and deflection under braking load, then peak rotor temperature and heat distribution between braking events to rule out fade and warping.
- Planned the hydraulic routing and fittings BOM; machined, drilled and lathe-finished the rotors, installed the calipers and master cylinders, and bled the system.



Rear rotor CAD — slotting and vane pattern set by thermal mass and inertia.

RESULT A fully modelled, simulated and manufactured brake system. The simulation work also left the team a framework for evaluating rotor size, master cylinder and bias changes on future cars.



Thermal simulation of the rear rotor at t = 20 s of braking.



Machining rotor features on the mill.



Rotor and upright integrated on the car.

TOOLS SolidWorks · ANSYS (structural & thermal) · MATLAB · Mill, lathe & surface grinder · Hydraulic system integration

Water Guard

Class project for a major Chicago aquarium · team of five

A removable, reusable waterproofing system that protects a ~$1M portable CT scanner during aquatic animal imaging — replacing 45 minutes of plastic sheeting and painters tape with a pre-formed liner that installs in under four.

45 → 4 min	~$1M	12.5°	106 N
SETUP TIME	ASSET PROTECTED	RUNOFF SLOPE	WELD STRENGTH, WET

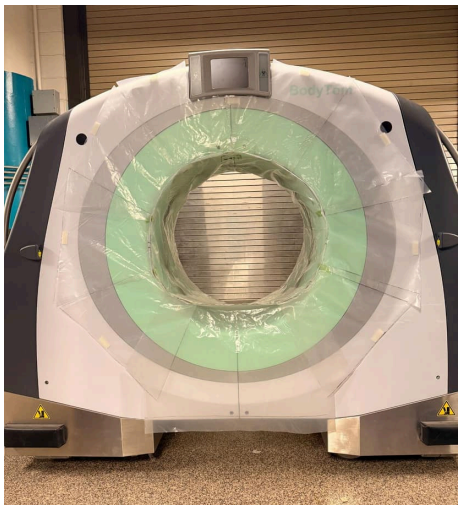
WHAT I DID

- Led the gantry liner geometry — an oblique frustum that turns the bore into a sloped surface draining to the rear, sized to hold a 12.5° slope inside the 6″ clearance beneath the gantry.
- Designed the overlapping seam strategy from tensile data and ran the saltwater degradation study on heat-welded LDPE (n = 5 per condition, 24 h at 1.025 g/cm³): 94.8 N dry against 106.4 N after exposure, failing cohesively rather than at the weld.
- Validated the runoff slope on a full-scale LDPE mockup, designing to the wrinkled-liner case rather than the taut one because the liner will wrinkle in storage.
- Synthesized NeuroLogica engineering interviews into hard design constraints, and contributed the FMEA that ranked seam integrity, attachment and runoff as the controlling risks.
- Added a clear PETG laser window after on-site testing showed the liner diffused the positioning laser past 15–20 cm — staff required it fully visible.

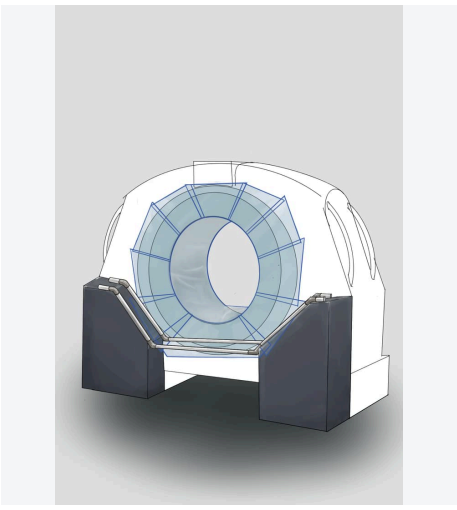


Fitting the gantry liner to the scanner on site.

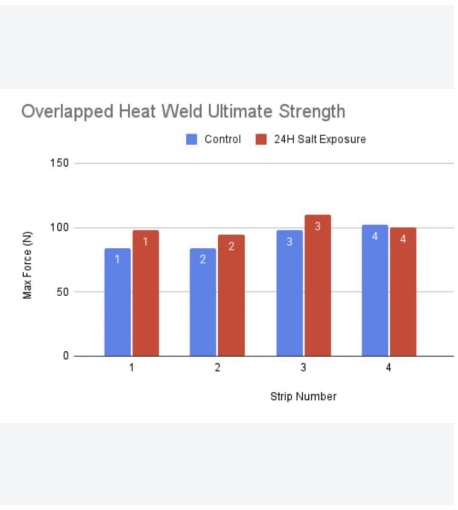
RESULT A full-scale prototype — sloped liner, front and back faces, laser window and runoff collector — handed over ready for use, with installation and field-repair instructions.



The finished liner installed on the scanner.



Liner architecture: gantry frustum, faces and rear collection.



LDPE weld strength, dry control against 24 h saltwater exposure.

TOOLS Polymer materials testing · Heat welding · Tensile & leak testing · FMEA · Stakeholder interviews · Design for serviceability

Smart Sheet Smith

Advanced Intelligent Manufacturing Lab, Northwestern · NSF ERC-HAMMER (EEC-2133630)

A five-agent system that turns a 2D sheet-metal drawing into a verified, physics-grounded bending process plan — no 3D CAD model and no labeled training data. Co-authored and presented at MSEC 2026 / NAMRC54.

5	50	2	100s
SPECIALIZED AGENTS	INDUSTRIAL DRAWINGS	SELF-CORRECTING LOOPS	DRAWINGS AUTO-GENERATED

Vision layout & thickness → **Tracer** geometry → JSON ⇄ **Geometric Judge** reference-free → **Engineering** physics via RAG ⇄ **Performance Judge** plan check → **Flat-pattern blueprint**

WHAT I DID

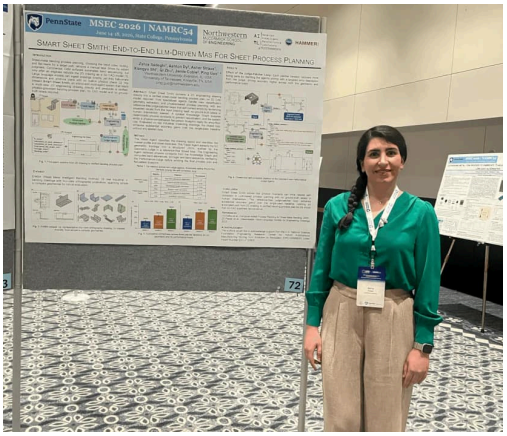
Worked on the Tracer Agent — the stage that extracts a drawing's full geometric topology into structured JSON — and owned much of the data foundation it was developed and validated against.

Modelled the parts in CAD, produced the multi-view orthographic drawings, hand-derived the ground truth, and did the bend math — bend allowances, developed lengths, segment sequences — case by case.

Applied a manufacturability check the math alone cannot: reasoning through the forming process to judge whether a plan was actually realizable on a press brake. That drove several corrections to bend ordering and tool access.

Reviewed Engineering Agent output as the manufacturing engineer in the loop — found bend deduction computed from the outer instead of the inner radius, a K-factor that had to be fixed before deduction, and V-die selection reported as one number rather than a realistic 8t–10t range.

Built a SolidWorks VBA pipeline that randomizes a base model's dimensions and exports fully-dimensioned drawings automatically, with a rebuild and "safe box" check that retries failures instead of emitting bad samples — cutting dataset creation from days to minutes at an 85–90% yield across three part families.



Our poster at MSEC 2026 / NAMRC54, presented by Zahra Sadeghi, the graduate student I worked with.

RESULT The reference-free judge–patcher loop lifted accuracy substantially over the single-pass baseline without a single labeled example — an automated path from a 2D print to a verified bending plan for job shops that 3D-CAD pipelines cannot serve.

THE SIX CHECKS THAT DRIVE SELF-CORRECTION

SSA Segment sequence alignment — GEOMETRIC JUDGE	**RAA** Feature anchor accuracy — GEOMETRIC JUDGE	**GDD** Developed length deviation — GEOMETRIC JUDGE
PMA Physics math accuracy — PERFORMANCE JUDGE	**OSA** Operation sequence alignment — PERFORMANCE JUDGE	**FVA** Feasibility validation — PERFORMANCE JUDGE

Each judge re-derives the expected value from the input drawing itself rather than from an answer key, so a failed check can rewrite the upstream agent's prompt and re-run that stage. Evaluated on SIMBA — 50 real industrial V-bending drawings with four-view orthographic projections, spanning simple through complex geometries.

TOOLS Multi-agent LLM orchestration · Vision-language models · RAG · SolidWorks API / VBA · Sheet-metal bend analysis · Dataset construction

Performance Bicycle Brake Caliper

Mechanical Design & Manufacturing · Northwestern

Designed, simulated, printed and road-tested a topology-optimized bicycle brake caliper against ISO 4210 — then proved with a second iteration that the lighter part was the worse engineering answer.

41.3 g	11.3 m	68.9 N	$8.26
FINAL MASS	STOPPING DISTANCE	RIDER INPUT FORCE	COST PER PART

WHAT I DID

Translated qualitative user needs into quantified targets drawn from ISO 4210 and benchmarks: stop a 150 lb rider within 15 m from 10 mph, at or under 70 N input, with tip deflection at or under 11 mm and 500,000 load cycles.

Ran FEA across the full allowable design envelope to find where the material was doing no work, then used topology optimization to pull it onto the load paths.

Printed in SLS Nylon-12 and tested on a bicycle: 41.3 g, 11.3 m stop, 68.9 N input, 3 mm and 5 mm arm deflection — every critical metric passed, with the calipers locking the wheels.

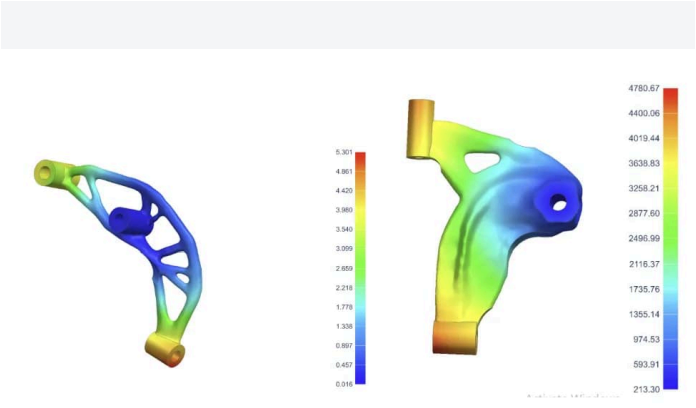
Iteration 2 cut 14% of the mass through larger cutouts and wider pivots. The lost stiffness raised required input force 48% to 102 N, breaking the requirement — so iteration 1 stayed.

Analyzed manufacturing across three volumes: die casting at 50k/yr, forging plus CNC at 10k/yr, and additive at 200/yr for custom builds.



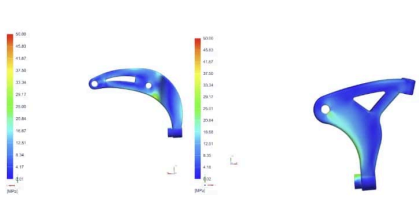
The SLS Nylon-12 caliper, printed from the optimized geometry.

RESULT All safety metrics met, and a documented stiffness–weight tradeoff: six grams of mass saved cost 48% more rider input force. The heavier design was the right one.

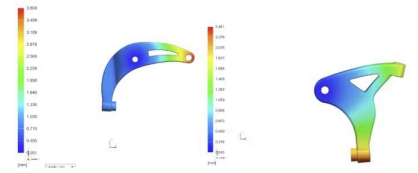


Figures 5a, 5b: Initial Topology Optimization on Design Spaces for Left and Right Calipers

Topology optimization resolving the load paths into rib-like structure.



Figures 7a, 7b: FEA on First Printed Design for Left and Right Calipers (Stress)



Figures 7c, 7d: FEA on First Printed Design for Left and Right Calipers (Displacement)

FEA on the first design: stress and displacement, safety factor ≈ 2.75 on deflection.

TOOLS FEA · Topology optimization · ISO 4210 · SLS Nylon-12 · Experimental validation · DFM at volume

ExtendIt

Human-Centered Product Design · Northwestern

A permanently installed hinged desk extender that adds 81 sq. in. of usable lecture-hall workspace, folds flush beneath the desk when stored, and was preferred by 8 of 10 users over four alternative concepts.

+81 in²	8 / 10	33.5 lb	$10.19
ADDED WORKSPACE	USER PREFERENCE	TESTED LOAD	COST PER UNIT

WHAT I DID

- Ran 22 exploratory user interviews and ranked the opportunity spaces on a weighted scoring matrix before committing to a direction.
- Tested seven physical mockups with ten users each; students consistently chose a permanent, stable fixture over portable clip-on attachments.
- Contributed the mechanical design and hinge selection — a galvanized locking hinge with the 90° detent ground off, so it locks only at 0° stored and 180° deployed.
- Built a dedicated load-test rig to avoid damaging the prototype: a 33.5 lb suspended load gave ⅜″ deflection 6″ from the hinge, inside the ≤0.5″ target, with 9.5″ of seating clearance when folded.
- Modelled unit economics at $10.19 — $6.02 materials and $4.17 of labor — against a U.S. school furniture market above $2B, with B2B university sales and licensing as the route to scale.



The final prototype deployed on a lecture hall chair.

RESULT A prototype that exceeded its strength requirement and won head-to-head user preference, with the hinge, clearance and cost structure all resolved for a permanent installation.



Load and deflection testing on the dedicated rig.



Simulated real-world loading during user testing.



The modified locking hinge installed beneath the desk.

TOOLS User research & interviews · Mechanical design · Structural load testing · Rapid prototyping · Cost modeling

Hands-On Capability

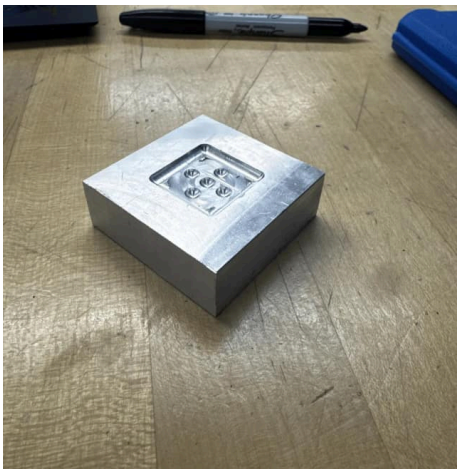
I make the parts I draw. Across coursework, research and club work I have machined, brazed, formed, cast, printed and waterjet-cut my own designs — and that is most of why they come out manufacturable. A sample of the work behind the projects in this book.



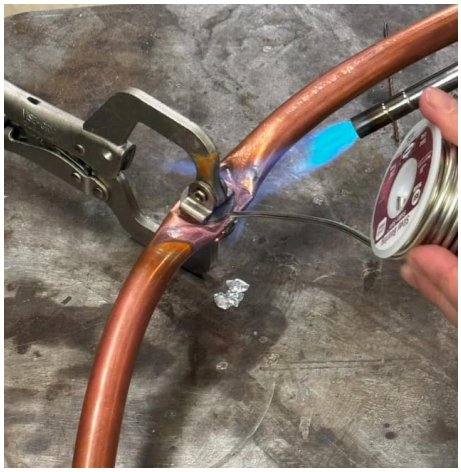
Assembling a mechanism at the bench



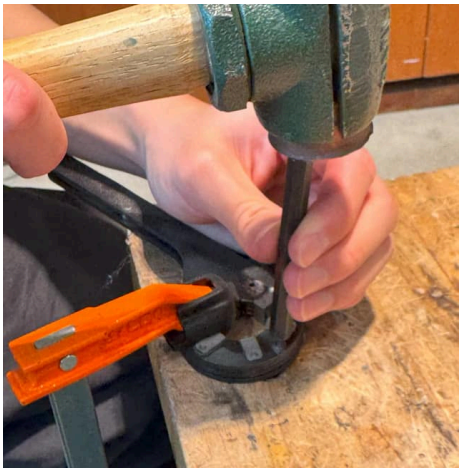
Machining rotor features on the mill



CNC-machined aluminum mold



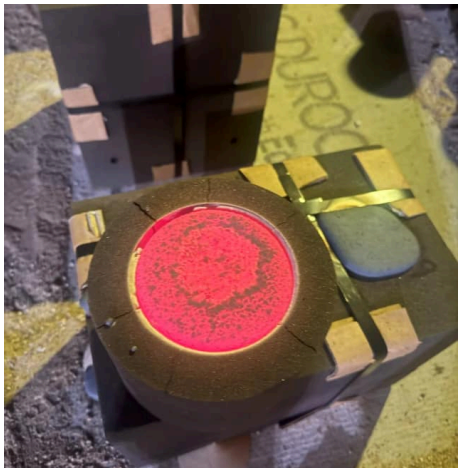
Torch-brazing a copper joint



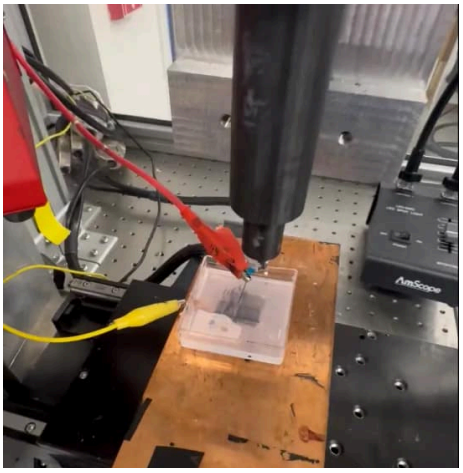
Hand-forming metal



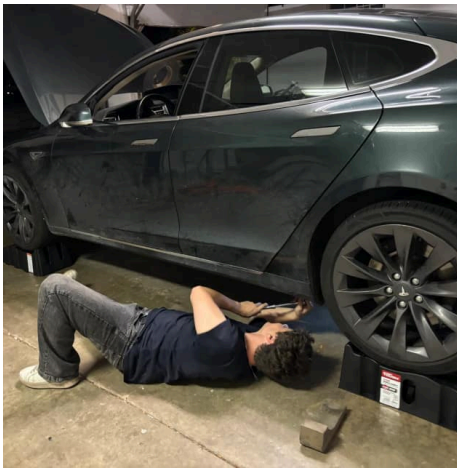
Abrasive waterjet cutting



Sand mold during an iron pour



Near-field electrospinning at 10 kV, AIM Lab



Under a Model S at Jordan's Tesla Repair

TOOLS Mill, lathe & CNC · Waterjet · Laser cutting · FDM, SLS & PETG printing · Welding & brazing · Sand and iron casting · Sheet metal · Metrology

More Work

Each of these has a full write-up — process, drawings, test data and photographs — on the portfolio site, along with a gallery of fabrication and studio work.

Flexible Tactile Sensors

AIM Lab · 2025

Hand-fabricated more than ten flexible resistive sensors — carbon nanotubes and nanoparticles embedded in Ecoflex silicone — for a soft robotic hand, and ran the near-field electrospinning rig at 10 kV.

Midwest EV — Openpilot Retrofit

2026 · in progress

Enclosure design for the custom PCB in an openpilot retrofit for Tesla Model S cars built before factory Autopilot, iterating as the board moves through design review.

SME Dorm Accessibility Project

2026 – 2027 · in progress

Leading a team of five in SME's National Student Manufacturing Innovation Challenge, designing for residence-hall accessibility.

Kerf-Bent Walnut Turntable Stand

2026 · in progress

An S-curved kerf-bent walnut stand with three platforms and brass accents, tuned for vibration isolation; CAD and BOM finalized, awaiting CNC router time.

Drivetrain Efficiency Test Rig

Baja SAE · 2025 – 2026

A bench-top rig that measures CVT and gearbox power losses separately under repeatable controlled load, instrumented with torque sensors and a Teensy 4.1 DAQ.

P1 Motor Club Track Model

2026 · in progress

A large-scale, mostly 3D-printed model of the club's racetrack property, built from the civil engineer's Civil 3D files and sized to travel in the back of a truck as a sales tool.

StimSpin & Patchwork Plush

Misericordia · 2024

Sensory devices for adults with cerebral palsy and developmental disabilities — a customizable spinning wheel and two tactile textile designs — developed with staff and tested with users.

Shop Projects

2026

Iron sand casting from a coke-fired furnace I built, a modular dodecahedron lamp, snap-fit PETG projector mounts, and a laser-cut acrylic light tracing board.



A finished flexible resistive sensor — conductive network embedded in Ecoflex.



The Baja SAE car the drivetrain efficiency rig is being built around.

FULL PORTFOLIO

strausasher.github.io/Portfolio

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